AP & RAP Assignment 1

1. Let $\Omega = \{1, 2, 3\}$, $\mathcal{F}_1 = \big\{\{1\}, \{2, 3\}, \Omega, \emptyset\big\}$ and $\mathcal{F}_2 = \big\{\{1, 2\}, \{3\}, \Omega, \emptyset\big\}$. Show that:
   (a) $\mathcal{F}_1$ and $\mathcal{F}_2$ are both σ-algebras.
   (b) $\mathcal{F}_1 \cup \mathcal{F}_2$ is not a σ-algebra.

2. Let $(X, \mathcal{A})$ be a measurable space, $Y \subset X$ and $\mathcal{A}|Y := \{A \cap Y : A \in \mathcal{A}\}$. Show that $\mathcal{A}|Y$ is a σ-algebra over Y.

3. Let $\Omega = \{1, 2, 3, 4, 5\}$ and $\mathcal{U} := \big\{\{1, 2, 3\}, \{3, 4, 5\}\big\}$. Determine $\sigma(\mathcal{U})$.

4. Let $\mathcal{B}(\mathbb{R})$ be the Borel σ-algebra over $\mathbb{R}$. Show that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$, where $\mathcal{C} = \{(c, \infty) \mid c \in \mathbb{R}\}$.